AI-Based Resume Screening System (CLI Application)

Submitted By

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INTRODUCTION

In today's digital world, recruitment plays a crucial role in organizational growth. However, manually reviewing a large number of resumes is time-consuming and inefficient. This project presents a Command Line Interface (CLI)-based system that automates resume screening using basic Artificial Intelligence techniques.

The system evaluates candidates based on their skills and experience and provides a ranking to assist decision-making. It is lightweight, easy to use, and runs entirely in a terminal environment.

MOTIVATION OF THE PROJECT

During observation of real-world hiring processes, it was noticed that recruiters spend a significant amount of time filtering resumes. Many deserving candidates may also get overlooked due to human limitations.

This motivated the development of an automated system that can assist in faster and more consistent candidate evaluation.

PROBLEM STATEMENT

Organizations face difficulty in efficiently screening and shortlisting candidates from a large pool of applications within limited time.

Manual screening:

- Takes too much time
- Is prone to human error
- Lacks consistency

OBJECTIVE OF THE PROJECT

- To automate the resume screening process
- To reduce manual effort in recruitment
- To apply AI concepts in a real-world scenario
- To build a complete CLI-based application

EXISTING METHODS

- Manual resume screening by HR
- Online Applicant Tracking Systems (ATS)

PROS AND CONS OF EXISTING METHODS

Pros:

- Human understanding of context
- Flexible decision-making

Cons:

- Time-consuming
- Inconsistent evaluation
- High workload

HARDWARE AND SOFTWARE REQUIREMENTS

Hardware:

- Computer/Laptop

Software:

- Python
- Command Prompt / Terminal
- VS Code

METHODOLOGY AND GOAL

The system follows a structured approach:

1. User logs into the system
2. Adds candidate resumes
3. Inputs job requirements
4. System analyzes resumes
5. Scores and ranks candidates

The goal is to create a simple yet effective AI-based screening tool.

FUNCTIONAL MODULES DESIGN AND ANALYSIS

- **Authentication Module** → Login/Register system
- **Resume Module** → Add, view, edit, delete resumes
- **AI Engine Module** → Score calculation and classification
- **Analysis Module** → Ranking candidates

ALGORITHM DEVELOPMENT

1. Take input from user
2. Clean and process skills
3. Match candidate skills with job requirements
4. Calculate score
5. Classify candidate (Selected / Rejected)
6. Rank candidates

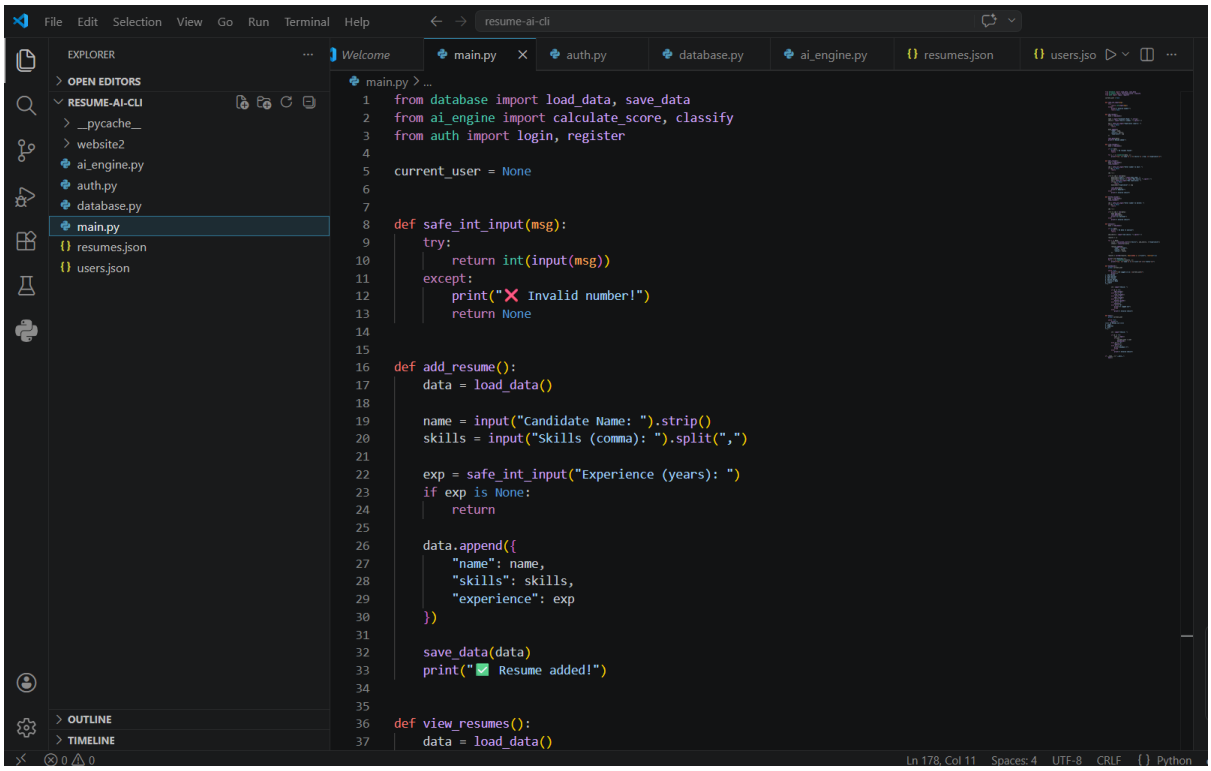
SOFTWARE ARCHITECTURAL DIAGRAM

User → CLI Interface → AI Engine → Data Storage → Output

CODING

The project is implemented using Python with modular structure:

- main.py → Main program
- auth.py → Login system
- database.py → Data handling
- ai_engine.py → AI logic

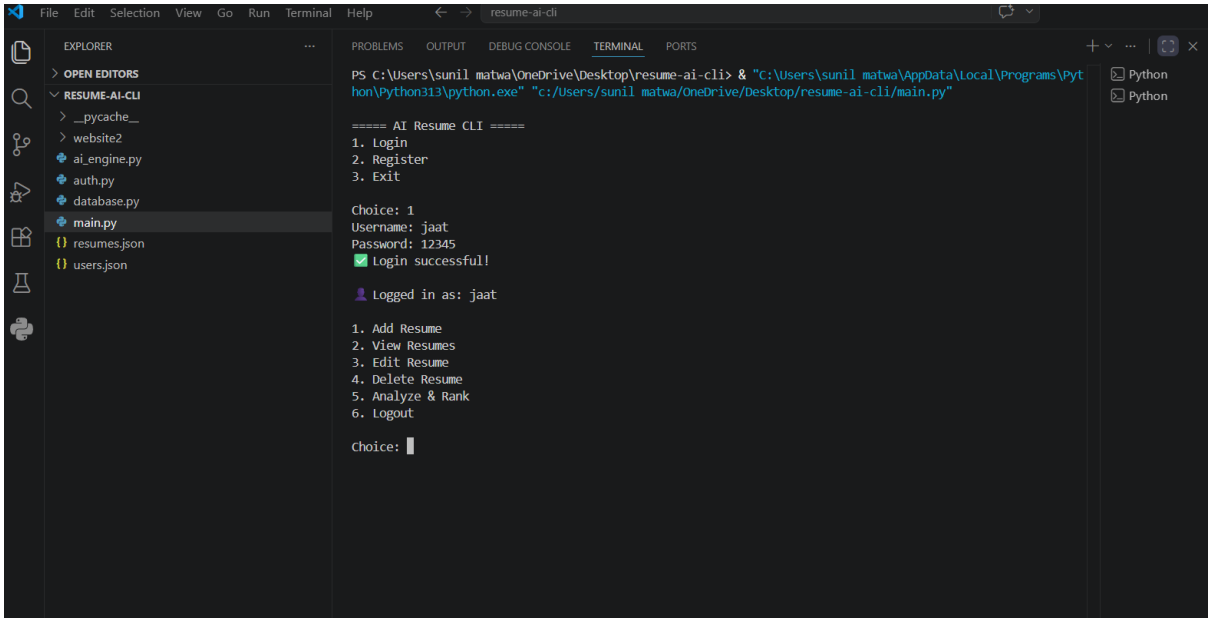


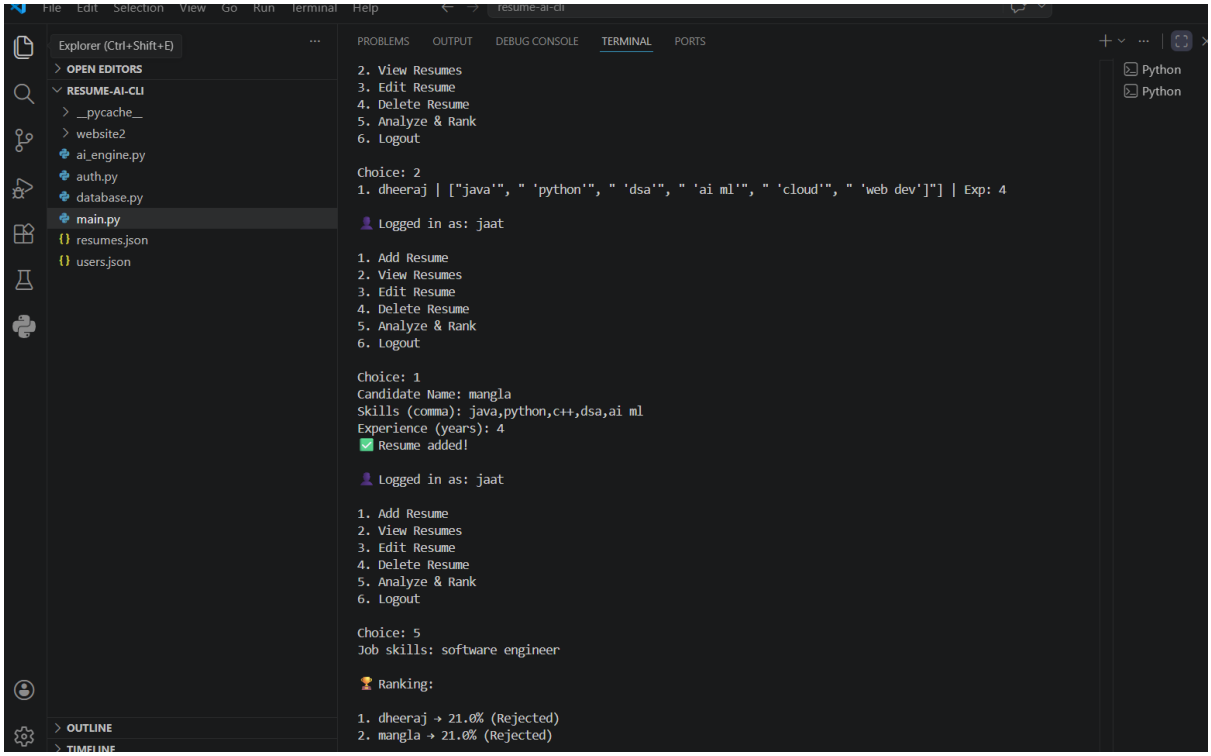
OUTPUT

Example:

1. dheeraj → 21% (rejected)

2. Mangla → 60% (Consider)




File Edit Selection View Go Run Terminal Help
resume-ai-cli
Explorer (Ctrl+Shift+E)
OPEN EDITORS
RESUME-AI-CLI
__pycache__
website2
ai_engine.py
auth.py
database.py
main.py
resumes.json
users.json
OUTLINE
TIMELINE
PROBLEMS OUTPUT DEBUG CONSOLE TERMINAL PORTS
Python
Python
2. View Resumes
3. Edit Resume
4. Delete Resume
5. Analyze & Rank
6. Logout
Choice: 2
1. dheeraj | ["java'", " 'python'", " 'dsa'", " 'ai ml'", " 'cloud'", " 'web dev']"] | Exp: 4
Logged in as: jaat
1. Add Resume
2. View Resumes
3. Edit Resume
4. Delete Resume
5. Analyze & Rank
6. Logout
Choice: 1
Candidate Name: mangla
Skills (comma): java,python,c++,dsa,ai ml
Experience (years): 4
Resume added!
Logged in as: jaat
1. Add Resume
2. View Resumes
3. Edit Resume
4. Delete Resume
5. Analyze & Rank
6. Logout
Choice: 5
Job skills: software engineer
Ranking:
1. dheeraj → 21.0% (Rejected)
2. mangla → 21.0% (Rejected)

KEY IMPLEMENTATIONS OUTLINE OF THE SYSTEM

- Secure login system
- Resume CRUD operations
- AI-based scoring algorithm
- Ranking system

SIGNIFICANT PROJECT OUTCOMES

- Reduced manual effort
- Faster decision-making
- Consistent evaluation

TESTING AND REFINEMENT

The system was tested with:

- Valid inputs
- Invalid inputs
- Empty data

Errors were handled using input validation.

PROJECT APPLICABILITY ON REAL-WORLD APPLICATIONS

- Recruitment systems
- HR tools
- Job portals

CONTRIBUTION / FINDINGS OF THE PROJECT

The project demonstrates how simple AI techniques can solve real-world problems effectively without complex models.

LIMITATIONS / CONSTRAINTS OF THE SYSTEM

- Uses basic AI logic
- No deep learning model
- Limited dataset

CONCLUSIONS

This project successfully implements an AI-based resume screening system using CLI. It provides a simple, efficient, and practical solution for recruitment challenges.

FUTURE ENHANCEMENTS

- Integration with Machine Learning models
- GUI-based interface
- Resume parsing from PDF
- Database integration

REFERENCES

- Python Documentation
- AI and ML Course Materials